

INTRODUCING CDS INDEX OPTIONS

Global Credit Derivative and Index Research

Dmytro Shelukhin ^{AC}

dmytro.shelukhin@jpmorgan.com

J.P. Morgan Securities Plc

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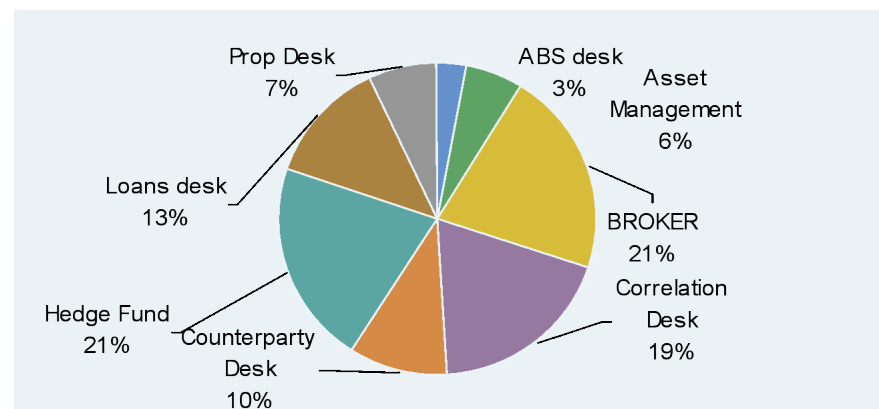
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Market Structure

- CDS options are popular non-linear instruments used to express outright views on spreads and volatility or to hedge existing positions and portfolios.
- €~10tn annual options volume, \$40bn a day
- 80% client trades
- Main bid/offer = 2-3c
- Typical size = €250m to €1bn

Trades by Counterparty



Source: J.P. Morgan

Account Type	Use of Options	Type of trade	Standard Size	Frequency
Correlation desks	Hedging gamma exposure.	Buy naked options or trade structure (payer spreads, ladders, cylinders).	500-2bn	Monthly
ABS counterparties	Hedging portfolios and options found to be a better instrument to hedge their underlying than the index – need <i>tail</i> .	Buyer of naked options.	250-500m	Monthly
Loan desk	Portfolio protection for large moves wider.	Buyer of naked options.	500-1bn	Fortnightly
Counterparty desks	Need gamma/tail.	Buyer of naked options, sometimes buy payers and sell receivers.	250-1bn	Monthly – Quarterly
HF / Prop	Anything	Gamma trades; cross market trades, portfolio hedges.	250-2bn	Every other day
Real Money	Hedging & alpha generation (alt risk premia)	Buyer of naked options; selling vol.	250-1bn	Monthly

Source: J. P. Morgan.

CDS Index Options

Option characteristics

- Option to buy/sell CDS index protection at a future date at a pre-agreed spread (strike).
- Options are traded on:
 - iTraxx Main
 - iTraxx Crossover
 - iTraxx Senior Financials
 - CDX.IG
 - CDX.HY
 - CDX.EM
- Traded on 5y index for all of the above with some liquidity in 10y index for iTraxx Main.
- Option expiries generally traded up until one year, though longer expiries may be available on request.
- Option expiries are on 3rd Wednesday of each month.
- CDS index options are **European-style** options; can only be exercised on expiry date between 9am and 4pm.
- Options can also be unwound/offset before expiry to take profit or cut losses.

Source: J.P. Morgan

CDS Index Options

CDS index option mechanics

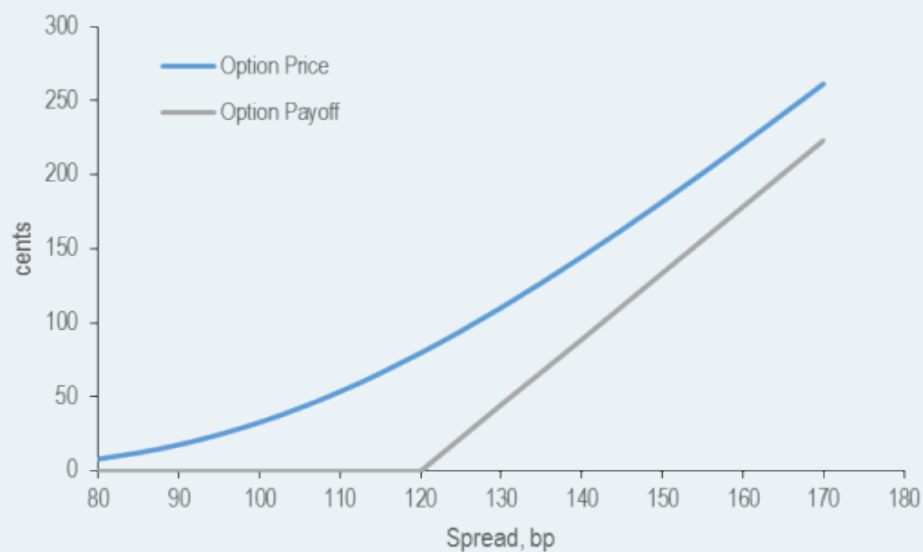
- Option types:
 - **Payer (put):** Option to buy protection, i.e. pay coupons
 - **Receiver (call):** Option to sell protection, i.e. receive coupons
- The option buyer pays an upfront **premium** which is usually denominated in cents; 1 cent = 0.01% on an upfront basis.
- If option is exercised, contract is settled by **physical delivery**
 - i.e. a new index trade is recorded and an upfront (calculated from the option strike) is exchanged
- **Example:**
 - Client buys iTraxx Main payer option with 3 months to expiry and strike of 70bp for 20c on notional of €100mm.
 - Initial payment is equal to $\text{€}100\text{mm} \times 20/10,000 = \text{€}200\text{k}$.
 - At expiry iTraxx Main spread is 80bp.
 - Client exercises option and buys protection from JPM at 70bp. This contract immediately has a MtM gain as the current index spread is wider than 70bp.
 - If the spread had been lower than 70bp, the contract would not have been exercised and would have expired worthless.

Source: J.P. Morgan

Payer Option Price & Payoff

iTraxx Main Payer Option

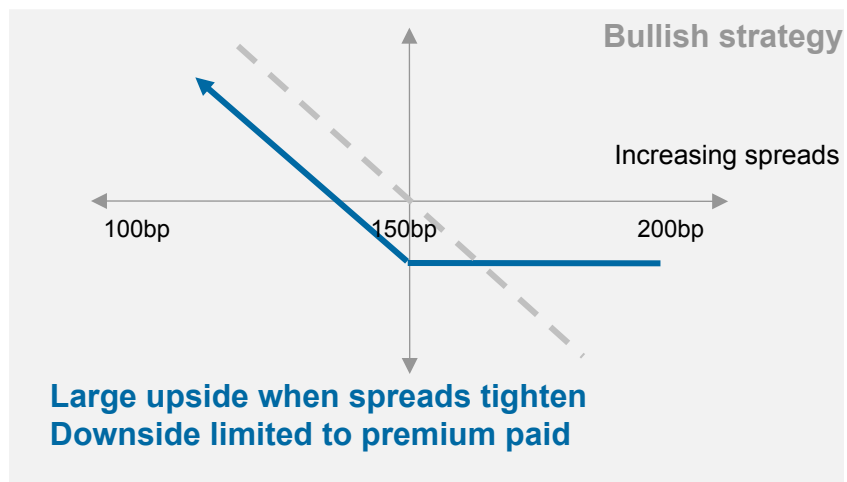
- Spot option price and payoff at expiry (before deducting premium) as a function of underlying index spread



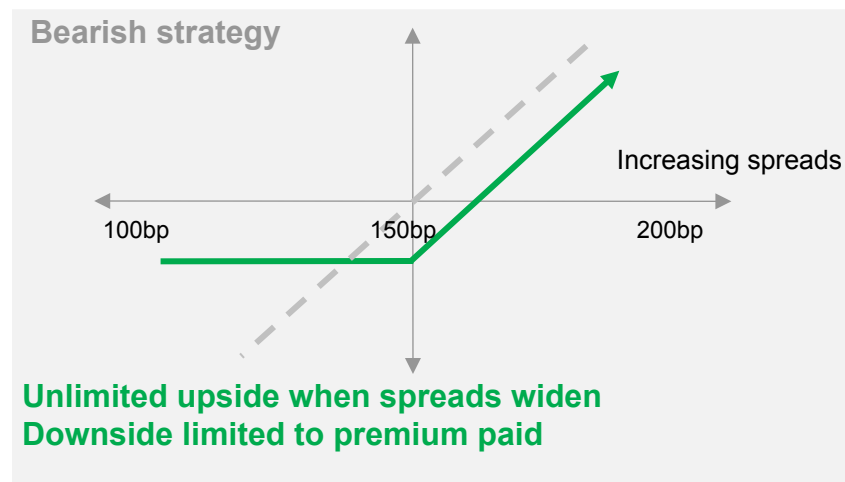
Source: J.P. Morgan

Receiver & Payer Options

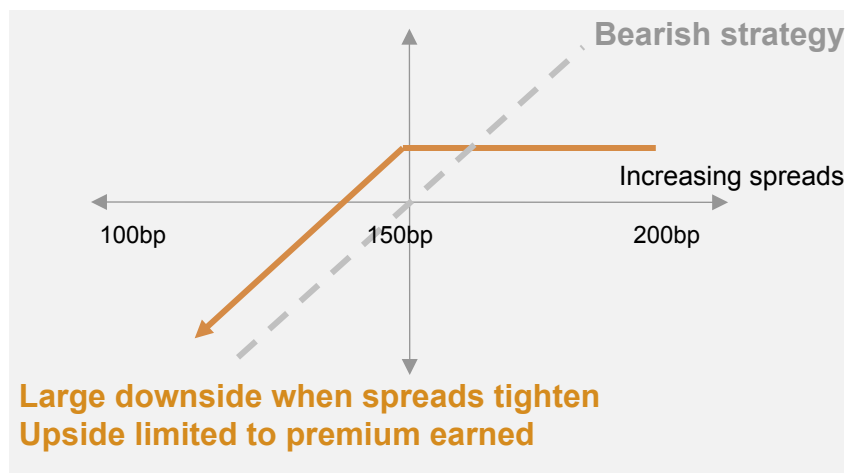
Buy receiver



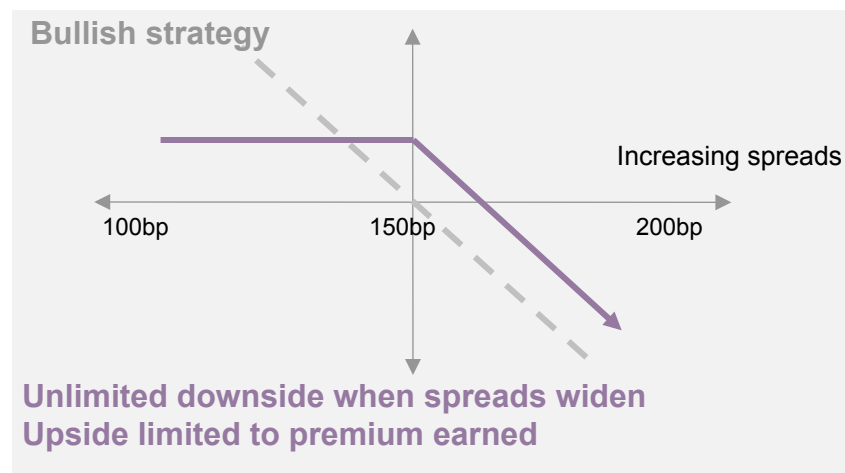
Buy payer



Sell receiver



Sell payer



Source: J. P. Morgan.

Example of a CDS Option Run

Underlying Index Prices (bid/offer) Expiry Forward Spread Current Spread

JPM iTrx Options: XOVER (S36V1) 15-Dec-21 ** Fwd @284.78, Delta @267.5

K	Rec	Delta	Vol	Chg	B/E	K	Pay	Delta	Vol	Chg	B/E
225	7.2/17.2	9%	41	2.7	6.2	250	185.1/195.1	76%	43	0.8	6.6
237.5	16.8/26.8	15%	41	2.1	6.3	262.5	150.5/160.5	65%	46	0.8	7.2
250	35.3/45.3	25%	43	0.8	6.7	275	124.0/134.0	56%	50	0.8	7.8
262.5	64.2/74.2	35%	46	0.8	7.3	287.5	103.9/113.9	48%	53	0.8	8.4
275	100.4/110.4	44%	50	0.8	7.9	300	88.6/98.6	42%	56	0.8	9.0
-	-	-	-	-	-	312.5	77.1/87.1	36%	60	0.8	9.6
-	-	-	-	-	-	325	68.4/78.4	32%	63	0.8	10.2
-	-	-	-	-	-	350	56.6/66.6	26%	69	0.8	11.4
-	-	-	-	-	-	375	47.5/57.5	21%	75	0.9	12.4
-	-	-	-	-	-	400	41.1/51.1	18%	80	0.9	13.3
-	-	-	-	-	-	425	33.9/43.9	15%	82	1.0	13.9
-	-	-	-	-	-	450	28.0/38.0	13%	85	1.0	14.3
-	-	-	-	-	-	475	22.9/32.9	11%	86	1.0	14.7
-	-	-	-	-	-	500	18.9/28.9	9%	88	1.0	15.0

Receiver/call options Implied volatilities (annual %) Implied volatilities (daily bp) Payer/put options Change on day

Strikes Deltas

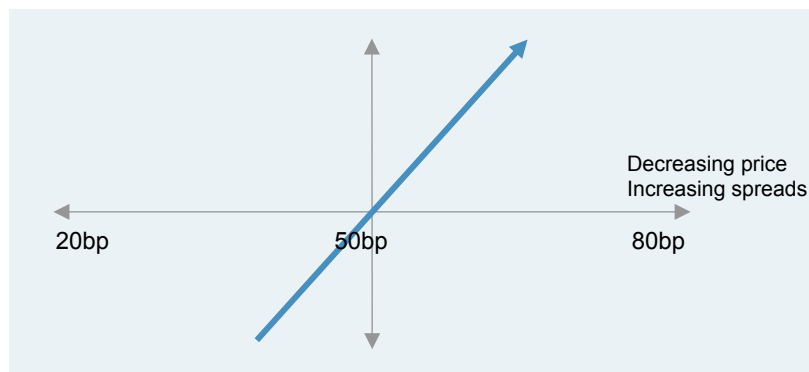
Source: J.P. Morgan, Bloomberg

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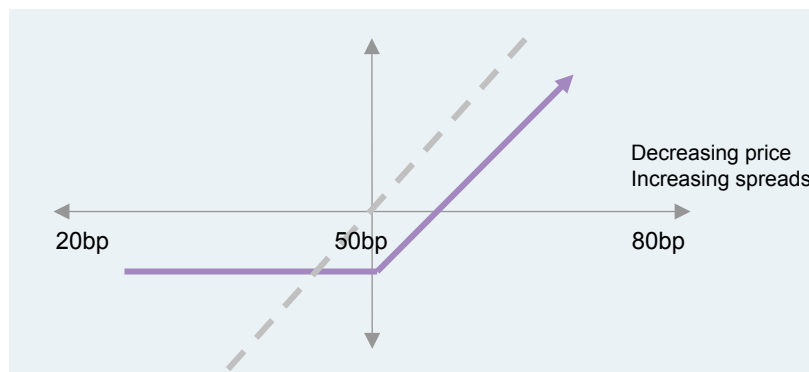
Expressing a bearish view on spreads

Buy index protection



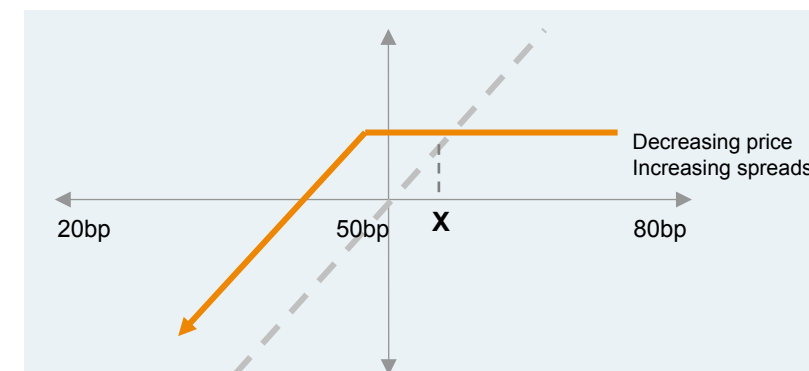
- Simplest strategy to take a view on spread widening
- Linear return profile if spreads widen or tighten
- Downside risk capped as spreads cannot be negative

Buy Payer



- Limit downside risk by buying a put option
- Full upside in spread widening (minus premium)
- Maximum loss limited to premium paid for option

Sell Receiver

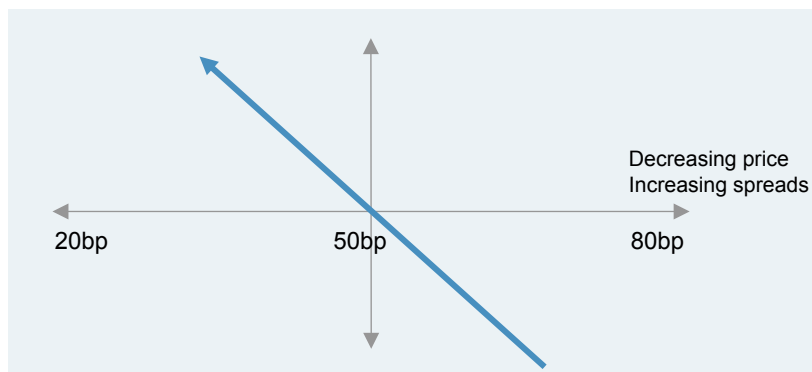


- Bullish view if believe spreads will widen but not past "x"
- Strategy outperforms buying index protection for levels below "x"

Source: J. P. Morgan.

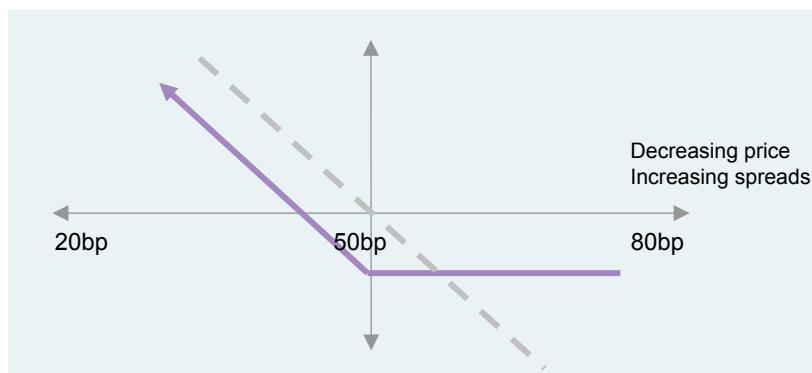
Expressing a bullish view on spreads

Sell index protection



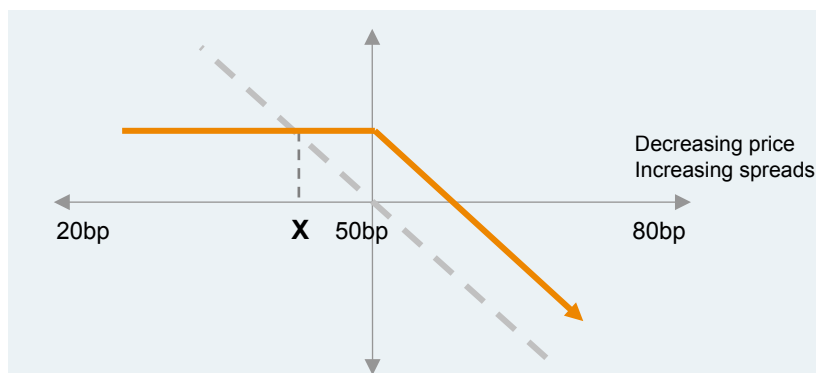
- Simplest strategy to take a view on spread tightening
- Linear return profile if spreads widen or tighten
- Unlimited downside risk if spreads widen

Buy Receiver



- Limit downside risk by buying a call option
- Full upside in spread tightening (minus premium)

Sell Payer



- Bullish view if believe spreads will tighten but not past "x"
- Strategy outperforms selling index protection for levels above "x"

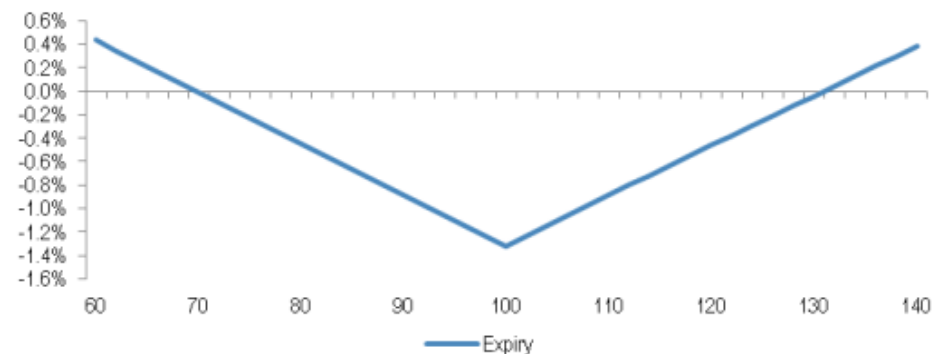
Source: J. P. Morgan.

Option Strategies – Range Strategies

Range bound strategies

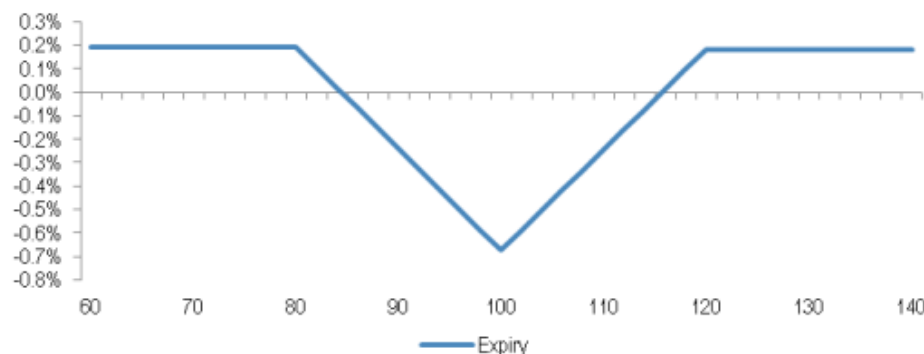
- **Straddle:** buy payer, buy receiver at same strike.
- **Strangle:** buy payer, buy receiver at different strike.
- **Butterfly:** buy straddle, sell strangle.
- Range bound option strategies profit if volatility goes up.

Straddles



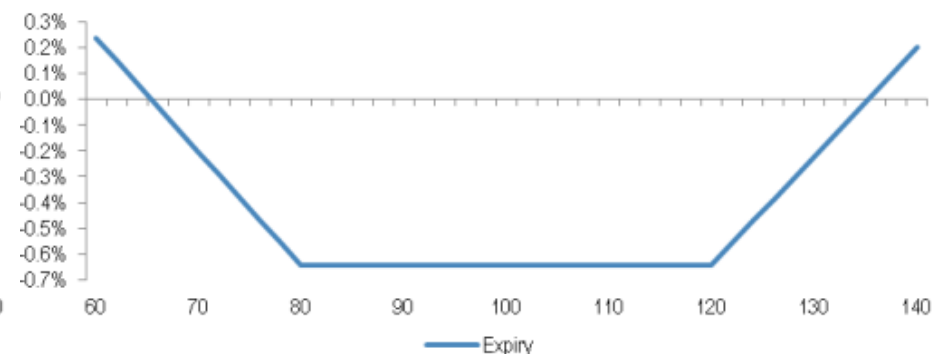
Source: J. P. Morgan

Butterflies



Source: J. P. Morgan

Strangles



Source: J. P. Morgan

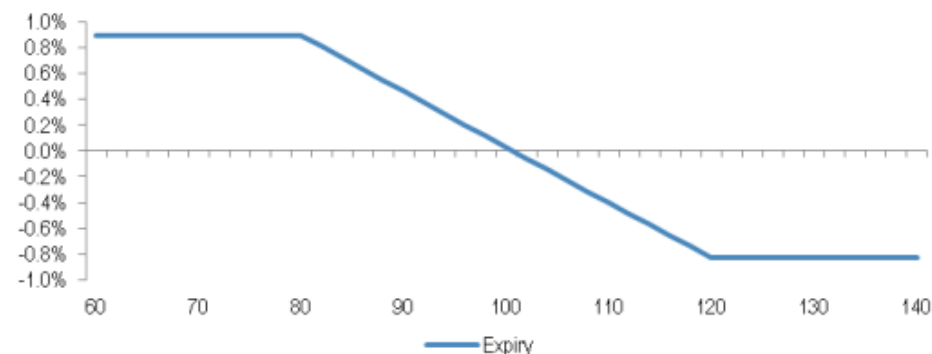
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Option Strategies – Bullish Strategies

Bullish strategies

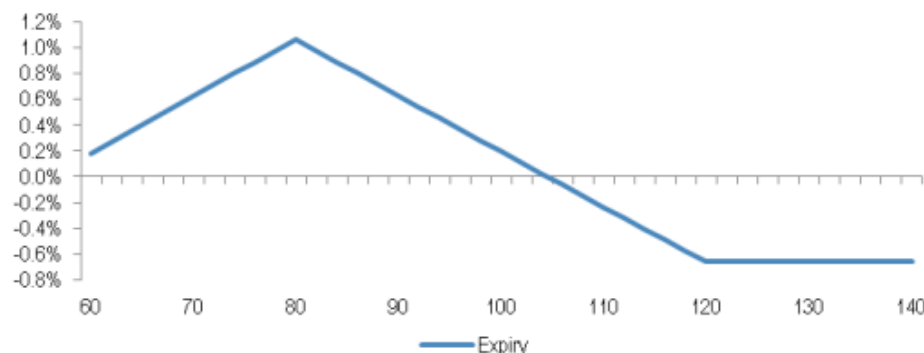
- **Receiver spread:** buy high strike receiver, sell low strike receiver. Benefits from flat skew.
- **Receiver ladder:** buy high strike receiver, sell 2 x low strike receiver. Benefits from flat skew.
- **Bullish risk reversal:** buy low strike receiver, sell high strike payer. Benefits from steep skew.

Receiver Spread



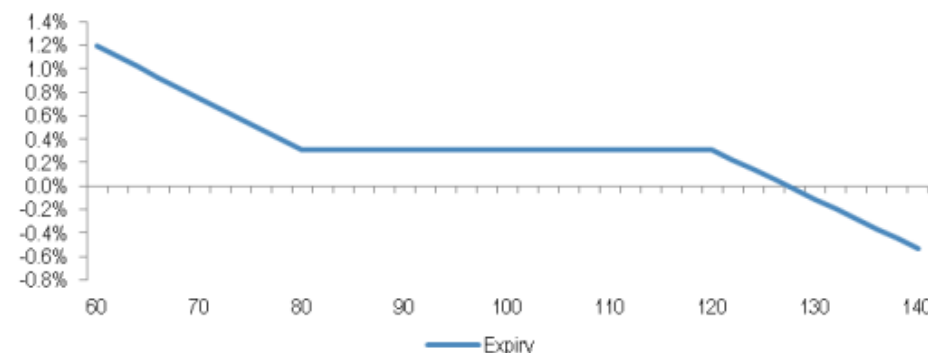
Source: J. P. Morgan

Receiver Ladder



Source: J. P. Morgan

Bullish risk reversal



Source: J. P. Morgan

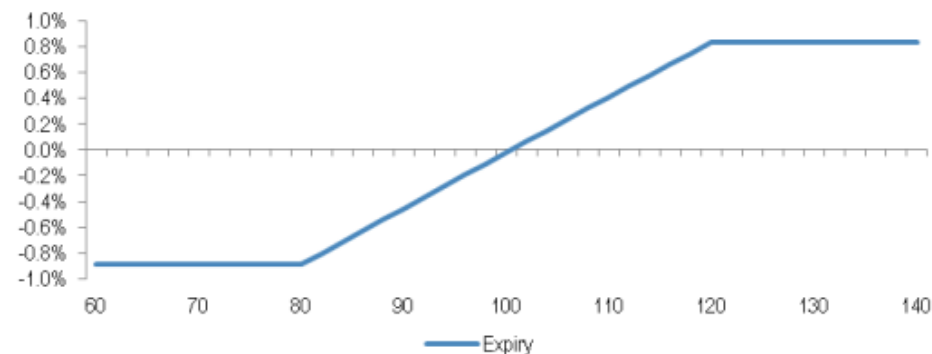
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Option Strategies – Bearish Strategies

Bearish strategies

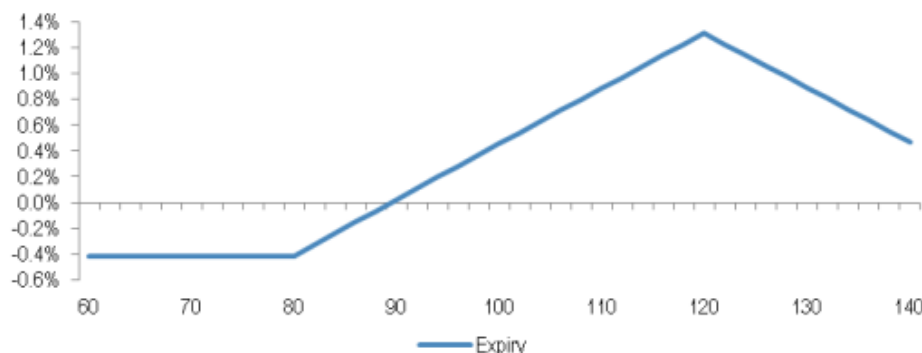
- **Payer spread:** buy low strike payer, sell high strike payer. Benefits from steep skew.
- **Payer ladder:** buy low strike payer, sell 2 x high strike payer. Benefits from steep skew.
- **Bearish risk reversal:** buy high strike payer, sell low strike receiver. Benefits from flat skew.

Payer Spread



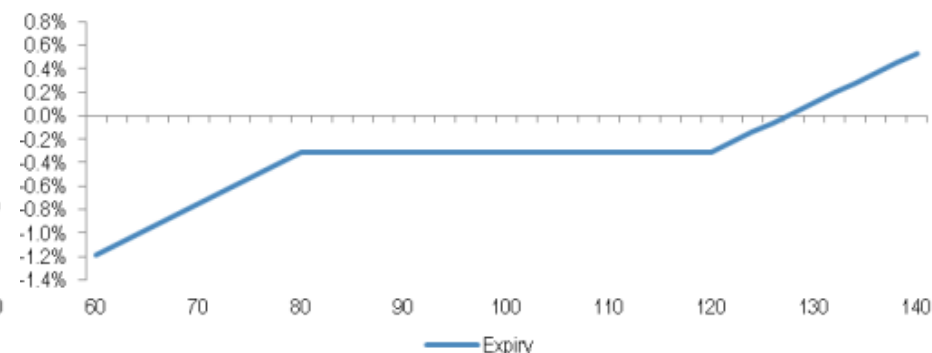
Source: J. P. Morgan

Payer Ladder



Source: J. P. Morgan

Bearish Risk Reversal



Source: J. P. Morgan

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CDS Option Questions

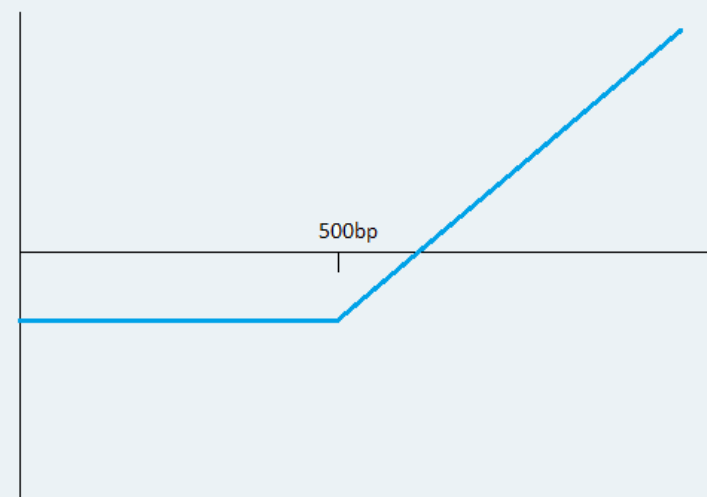
CDS index options

- How much do we pay for an option in cash terms?
- When will we exercise an option?
- What is the payoff for an option at expiry?
- At what spread level do we breakeven, taking the cost of the option into account?
- What happens if a name in the index defaults before expiry?

A simple example

Crossover 500bp option

- We buy a Crossover payer option with a 3-month expiry and a strike of 500bp on €100mm notional.
- Option cost = 150c
- Current spread = 420bp
- We have the right to buy iTraxx Crossover protection at 500bp at expiry. We will only do this if the spread on this date is greater than 500bp.
- **Initial Cost**
 - We pay 150c on €100mm notional
 - Upfront cost = €1.5mm

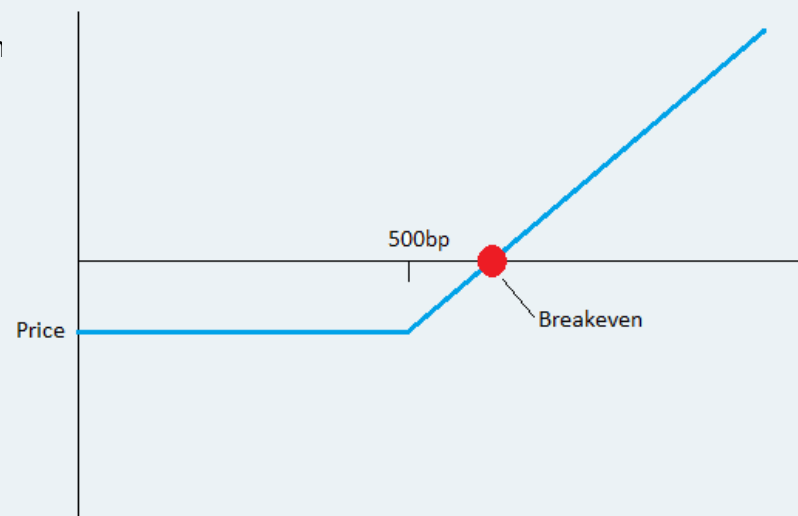


Source: J.P. Morgan

A simple example – breakevens

Crossover 500bp option

- If spreads are above 500bp at expiry, we will exercise our option and make a profit from selling index protection higher than the 500bp level at which we bought protection.
- However, we need to make back the cost we paid for the option - this means we need the index to widen to more than 500bp to make an overall profit.
- We need to calculate the **breakeven spread** – i.e. the spread at which the P&L from exercising the option is equal to the premium paid.
- For index, for a 1bp widening we make 1bp x Duration in profit.
- Breakeven spread = Strike + Option Cost / Forward Duration
- Forward duration = 3.5
Breakeven spread = 500bp + 150 / 3.5
Breakeven spread = 543bp
- This option will only make a total profit if spreads are wider than 543bp at expiry.



Source: J.P. Morgan

A simple example – P&L at expiry

Crossover 500bp option

- We can work out our **total P&L** at expiry in the same way we calculated the breakeven spread.
- If we exercise the payer option, we will buy protection at the strike K . We will then mark this protection to market at the current spread S .
- P&L if we exercise the option = $(\text{Spread} - \text{Strike}) \times \text{Forward Duration} \times \text{Notional} - \text{Option Cost}$
- P&L if we don't exercise the option = $-\text{Option Cost}$
- In previous example, we exercise the option when spreads are 600bp and the forward duration is 3.5
- P&L = $(600\text{bp} - 500\text{bp}) \times 3.5 \times \text{€}100,000,000 - \text{€}1,500,000$
= $\text{€}2,000,000$

Source: J.P. Morgan

Option worked examples

Option questions

- I buy an iTraxx Main receiver at a strike of 100bp with an expiry of 18-Sep for 30 cents on a notional of €200mm.

- What cash price do I pay for this option?

$$\begin{aligned}
 \text{Cash price} &= \text{Option cost} \times \text{Notional} \\
 &= 30c \times €200,000,000 \\
 &= €600,000
 \end{aligned}$$

- The forward duration of the option is 4. What is the breakeven of the option at expiry?

$$\begin{aligned}
 \text{Breakeven} &= \text{Strike} - \text{Option Cost} / \text{Forward Duration} \\
 &= 100bp - 30c / 4 \\
 &= 92.5bp
 \end{aligned}$$

- If the iTraxx Main spread is equal to 105bp at expiry, what is the P&L of this trade?

Spread is above strike so receiver option is not exercised. Total P&L of trade is just the loss of the option price.

$$\text{Trade P\&L} = -€600,000$$

- If the iTraxx Main spread is equal to 80bp at expiry, what is the P&L of the trade?

Spread is above strike so receiver option is exercised.

$$\begin{aligned}
 \text{P\&L} &= (\text{Strike} - \text{Spread}) \times \text{Forward Duration} \times \text{Notional} - \text{Option Cost} \\
 &= (100bp - 80bp) \times 4 \times €200,000,000 - €600,000 \\
 &= €1,000,000
 \end{aligned}$$

Source: J.P. Morgan

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What happens if there is a default in the index?

Defaults and CDS Index Options

- CDS **index** options trade with what is known as a “**no-knockout**” feature.
- This means that options give you protection on defaults before expiry.
- For example, if we buy a December payer option and there is a default in October, at expiry we receive the same default payout we would have received from an index position **provided we exercise the option**.
- In other words, options continue to reference V1 of the index after a default (with V2 index options also trading in parallel).
- **This means that the buyer of a payer option has default protection starting today, even though they are only buying protection starting on the exercise date** (often referred to as *front-end protection*).
- Example: we buy an iTraxx Crossover payer option with a 3m expiry on a notional of €150mm.
 - RiskyCo has a Failure To Pay credit event one month later with a recovery rate of 40%.
 - There are 75 names in the Crossover index, so Risky Corp makes up 1/75 (or 1.333%) of the index.
 - At expiry we exercise the option. We receive a payout of €1.2mm from the Risky Corp event ($= €150\text{mm}/75 \times (1-40\%)$) and also get entered into Version 2 of the index (i.e. index portfolio with remaining 74 names).
 - The effective notional of Crossover Version 2 protection is €148mm (i.e. the original notional minus that allocated to RiskyCo).

Source: J.P. Morgan

Forward spreads in options

CDS Option Forwards

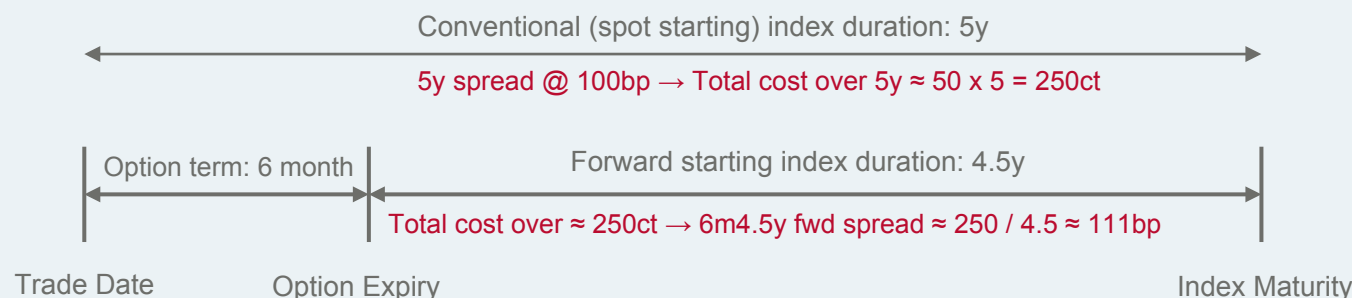
- We price CDS options using the forward rather than spot CDS spread (a la Black-76 model).
- This is the market-implied expectation for where spreads will be at the option expiry.
- The conventional CDS forward is calculated from the CDS curve; steeper curves mean higher forwards.
- **In options, the no-knockout convention means that payer options are worth more than they would be otherwise (vice versa for receivers).**
- This is incorporated into the option price by making an adjustment to the forward level.
 - If a name in the index defaults before the expiry of the option, we will be entered into an index with a defaulted name at expiry of the option. If we had bought a payer option, we could trigger the contract and collect on the defaulted name.
 - Therefore, we have received protection from today, even though the forward only offers protection from the option expiry.
 - We need to adjust the forward to account for the “No Knockout” feature of index options.
- The **Adjusted Forward** is approximately equal to the current spot spread multiplied by the ratio of the spot risky annuity (from today to index maturity) and the forward risky annuity (from option expiry to index maturity).

$$\text{Adjusted Forward} \approx S_{t,T} \times \frac{RA_{t,T}}{RA_{t,E,T}} = S_{t,T} \times \frac{RA_{t,T}}{RA_{t,T} - RA_{t,E}}$$

- **By convention, an at-the-money (ATM) option has a strike equal to the (adjusted) forward.**
- See *An Introduction to Options and Credit Volatility*, March 2007 for more details on CDS option forwards.

Forward spreads in options

Simple way of calculating CDS option forward



- CDS option underlying: *no-knockout* forward starting CDS protection.
- We are buying the same protection (from Trade Date to Index Maturity) but paying for it over a shorter period of time (from Option Expiry to Index Maturity), necessitating a higher running premium.
- Through a payer option, we own default protection for 5 years but only paying the running premium over 4.5 years. Total cost of protection should be (roughly) similar to the cost the spot-starting protection (we are simply re-distributing the cash-flows).
- Hence, the first order approximation for the forward spread is the total premium paid over 5y term, amortized over 4.5 years.
- The forward is therefore higher than the spot. The main reason for this is not because the curve is upward-sloping, but because it is a *no-knockout forward*.

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Option Valuation

CDS Option Valuation - Intuitively

- CDS index option prices are calculated using a Black-Scholes framework, modified for the CDS market.
- Pricing inputs:
 - Current spreads
 - Strike
 - Expiry
 - Rates
 - **Volatility**
- **Why is volatility important?**
- Options mean you get all the upside and none of the downside from market moves.
- Compare two different CDS indices, both of which have spreads of 100bp today.
 - Index A moves up or down by 1bp each day.
 - Index B moves up or down by 10bp each day.
- If we could buy a 100bp strike payer option on Index A or Index B for the same price, we would always buy the Index B option.
 - We don't care whether spreads tighten by 1bp or 10bp – our option is worthless in either case.
 - We make much more money if spreads widen by 10bp than 1bp.
 - We expect to make more money from a payer option on Index B; the price of this option will therefore be higher.

Source: J.P. Morgan

Implied and Realised Volatilities

CDS Options Volatilities

- CDS index options are quoted on a **price** basis.
- We can back out corresponding implied volatilities from the option prices.
 - Volatility is just another way to express option prices.
- By convention, implied volatilities are quoted on an annualised % of spread basis (annual spread vol).
- It is often more convenient to think in terms daily volatility in basis points (“bp vol” aka daily implied spread move) rather than annualised volatility in percentage terms.
- We can convert annualised volatility into daily volatility in basis points using the following formula:

- **Daily bp vol = % Annual Implied Vol x Index Spread / Sqrt(252)**

Annual Spread Vol = 50%
Index Spread = 100bp
Working Days = 252

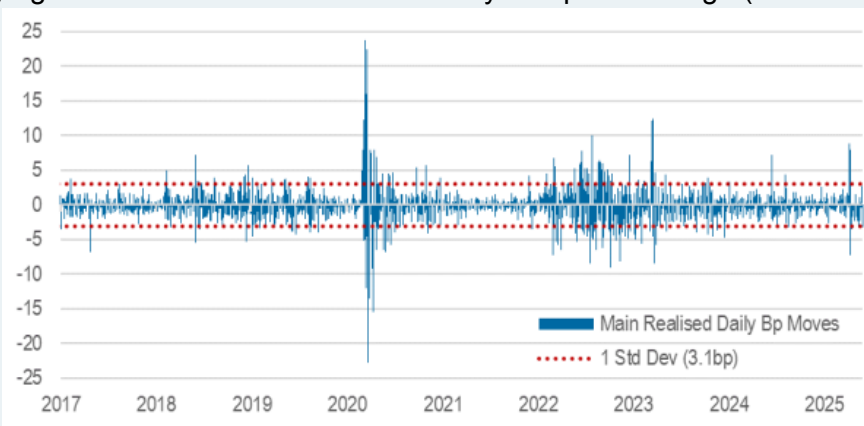
Relative to
absolute

Annual Implied Bp Move =
 $50\% \times 100\text{bp} = 50 \text{ bp/year}$

Annual to
daily

Daily Implied Bp Move =
 $50\text{bp} / \text{Sqrt}(252) = 3.1 \text{ bp/day}$

- The options market is implying that the index will move around by 3.1bp on average (1 std. dev.) every day.



Source: J. P. Morgan.

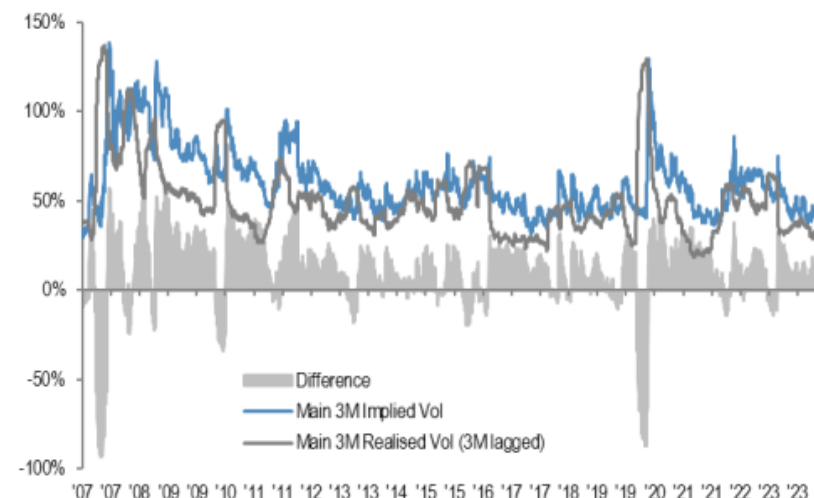
Historical Implied vs Realised Volatility

- To get an idea of whether these implied volatilities are cheap or expensive, we can compare them to historical realised volatilities.
 - **Historical Volatility** (aka realized or delivered vol) measures how variable spreads have been *in the past*. It is usually measured as the annualised standard deviation of daily percent or log changes of a variable over some historical period. It is **backward looking**.
 - **Implied Volatility** is option market-implied “forecast” of how variable the underlying spread is expected to be in the future (over the remaining life of an option). This forecast is calculated from an option premium as the level of volatility that gives a theoretical option price equal to its market price. It is model dependent and is **forward looking**.
- For example, if implied volatility was much higher than recent realised, this may be a good time to sell options.

Selling Credit Volatility

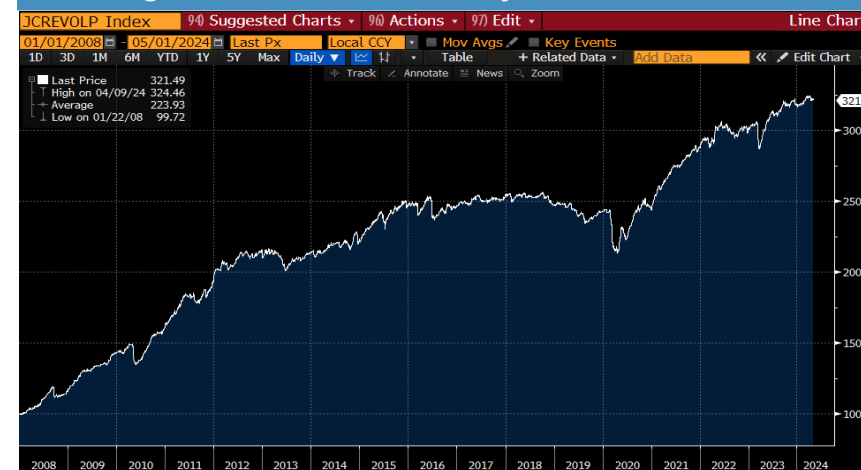
- Historically there were many more people looking to buy options than sell options in credit markets.
- This has historically driven implied volatility to be much higher than realised volatility.
- This difference can make it very attractive to sell volatility, even during periods of market stress.
- In rapid sell-offs (GFC in '08/'09, Covid-19), realised volatility can spike far beyond implied volatility.

Implied and Realised Volatility in iTraxx Main



Source: J. P. Morgan.

J.P. Morgan Global Credit Volatility Premium Index



Source: J. P. Morgan, Bloomberg

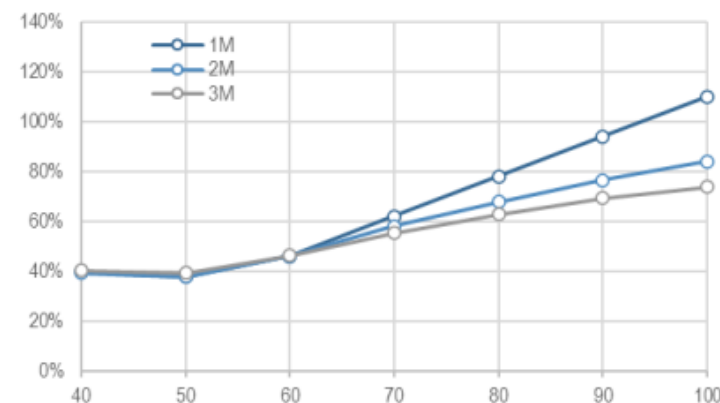
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Volatility Skew

Skew

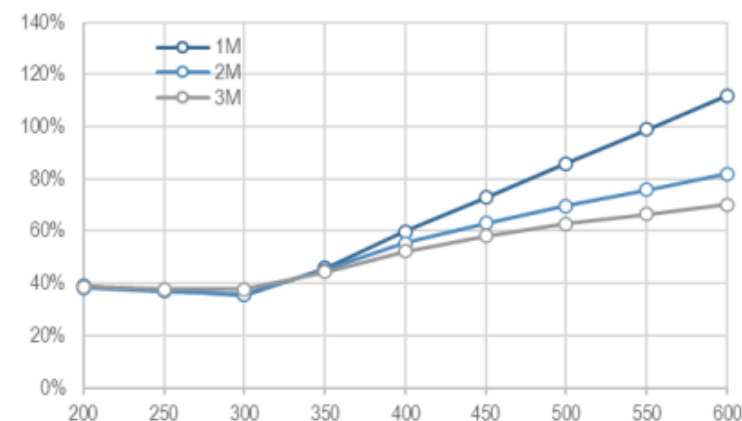
- Higher strike options typically trade with higher implied volatilities.
- This is a result of investors being happy to pay a premium for deep out-of-the-money options.
- Fundamentally can interpret the skew as being a result of the higher expected realised volatilities if spreads move wider.
- Steep skew makes certain trading strategies attractive – payer spreads/ladders and bullish risk reversals in particular.
- Trading skew allows investors to express a view on the market's perception of the probability of a large sell-off.
- More theoretically, vol skew can be interpreted as a signal that the market considers the log-normal assumption underlying the Black-Scholes model as inadequate. A positive skew suggests that the distribution of spread widenings is more heavy-tailed than a log-normal distribution.

iTraxx Main Volatility Surface



Source: J. P. Morgan.

iTraxx Xover Volatility Surface



Source: J. P. Morgan

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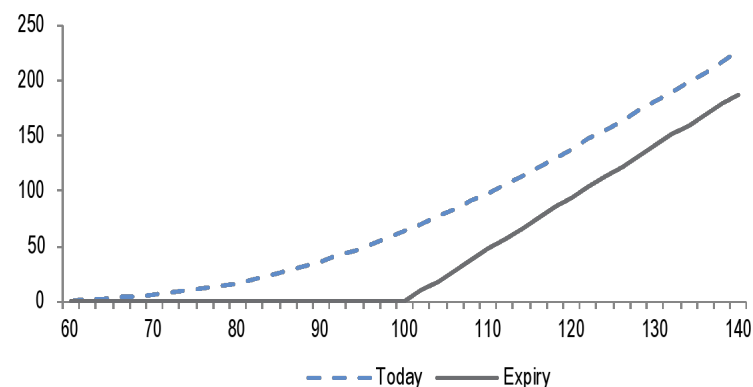
Options and Volatility

Trading volatility and delta-hedging

- Using an option pricing model we can value options prior to expiry.
- Investors used partial derivatives of the option price equation to keep track of how the price of the option is affected by changing markets.
- For most options, delta is the single biggest exposure. We can remove delta-exposure by trading a certain amount of index – this is known as **Delta Hedging**.
 - **Delta-Hedging** allows us to trade volatility outright.
 - The simplest way to create a long volatility position is to buy an ATM option (either payer or receiver) and buy or sell enough index protection to cancel out the delta of the option.
- Volatility positions seek to benefit from the following:
 - **Changes in implied volatility:** an increase in implied volatility benefits a long option position as the options become more valuable. This relies on the vega exposure of the position.
 - Long dated options are more driven by changes in implied volatility - **vega** P&L.
 - **Differences between implied and realised volatility over the trade horizon:** realised volatility that is higher than implied should benefit a long volatility position. This represents a difference in profit from gamma and theta exposures of a position.
 - Short dated options are more driven by differences in implied and realised volatility – **gamma vs. theta** P&L.

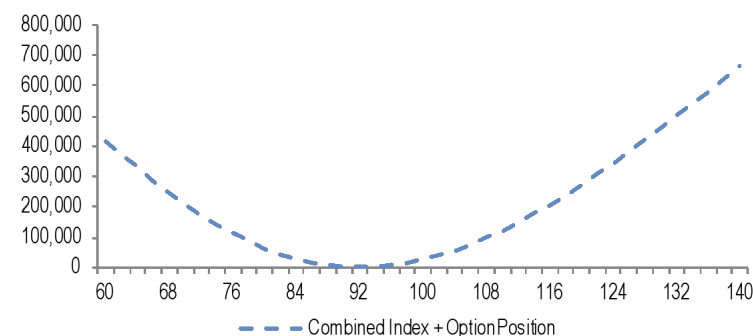
Source: J.P. Morgan

Option Value (ct) for iTraxx Main 100bp 3M Payer



Source: J. P. Morgan.

Delta Hedged Position – Index + Option



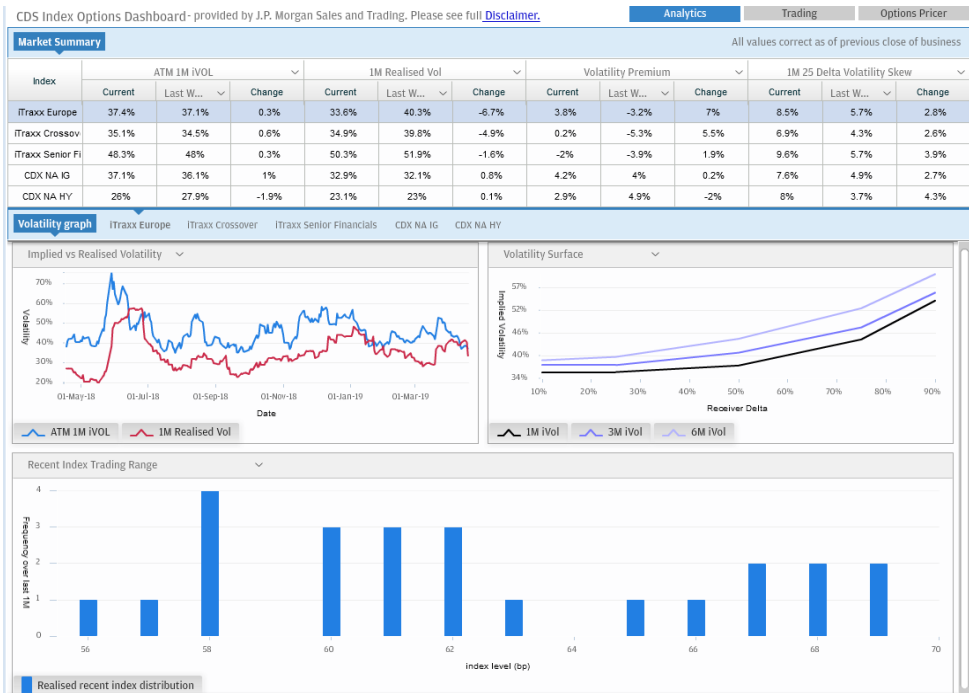
Source: J. P. Morgan.

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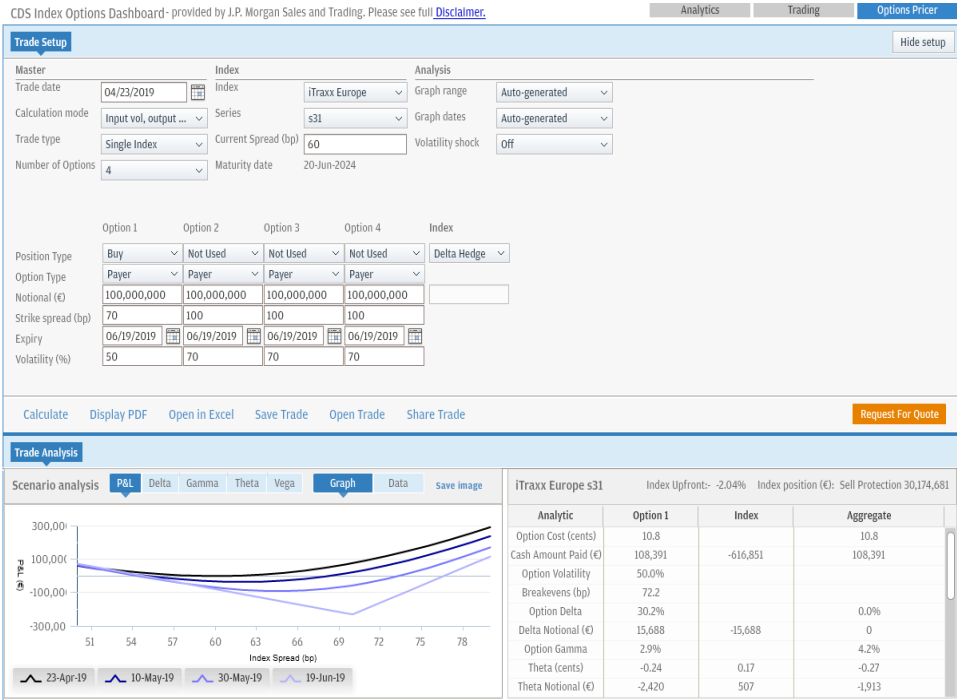
JPM Online CDS Options Trade Analyser

Web app on Morgan Market: https://markets.jpmorgan.com/#analytics.credit.cds_index_options

User guide: <https://www.jpmm.com/research/content/GPS-3689071-0>



Source: J. P. Morgan.



Source: J. P. Morgan.

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Additional Material

Additional Material

- *An Introduction to Credit Options and Credit Volatility*, 2007, J.P. Morgan
- *CDS Options Strategies*, 2011, J.P. Morgan
- *A CDS Option Miscellany*, 2019, R.J. Martin

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The impact of defaults on exercise breakevens

Defaults and CDS Index Options – 2

- This payout from defaults changes the economics of exercising the option.
- **Example:** an investor owns a €150mm payer option on iTraxx Crossover with a 3m expiry and a strike of 500bp.
- When the index has no defaults, the investor will be entered into a €150mm long protection position on the index should they exercise the option.
 - It is only economically beneficial for the investor to exercise the option when spreads are wider than 500bp.
- However, when there are defaults in the index **it can be economically beneficial for the investor to exercise the option even if the *new (V2)* index spread is below 500bp.**
- In the case of the Risky Corp default, if the investor exercises the option they will receive €1.2mm from the option described above and would be entered into a €148mm long protection position on version 2 of the Crossover index at 500bp.
 - P&L upon Exercise = P&L from defaults + MTM of index position
 - P&L upon Exercise $\approx$ P&L from defaults + Forward Duration $\times$ (Current Spread – Strike) $\times$ Reduced Notional
 - Setting *P&L upon Exercise* equal to zero and solving for *Current Spread* gives:
$$\text{Breakeven Exercise Spread}(bp) = \frac{- PL \text{ from Defaults} \times 10,000}{\text{Reduced Notional} \times \text{Forward Duration}} + \text{Strike}(bp)$$
- Using a Forward Duration of 3.5 gives a Breakeven Exercise Spread of 477bp in this case.
- It will be economically beneficial for the investor to exercise the option provided that the spread of *the version 2 of the index* is wider than 477bp.
- **The no-knockout convention means that payer options are worth more than they would be otherwise.**

Source: J.P. Morgan

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